

ATTACK IMPLEMENTATION AND DETECTION: IMPLEMENTATION GUIDE

1. Project Overview

This project demonstrates the deployment of a defensive security stack to detect specific MITRE ATT&CK techniques. By integrating Sysmon for deep endpoint visibility and Wazuh for centralized log analysis and alerting, we simulate and analyse sophisticated credential theft and lateral movement techniques in an isolated environment.

2. Lab Architecture

The lab was built using an isolated network to ensure safety.

- Wazuh Manager: Centralized engine for log collection and alerting.
- Windows Endpoint: Windows machine equipped with Sysmon and the Wazuh Agent.

3. Setup Steps

1. Wazuh Manager: Deployed in Virtual Box Manager.
2. Sysmon Installation: Installed on Windows 11 using sysmon64.exe

accepteula -i sysmonconfig.export.xml
3. Wazuh Agent: Installed on Windows and we have added this xml to

ossec config on windows

```
<localfile>
```

```
  <location>Microsoft-Windows-Sysmon/Operational</location>
```

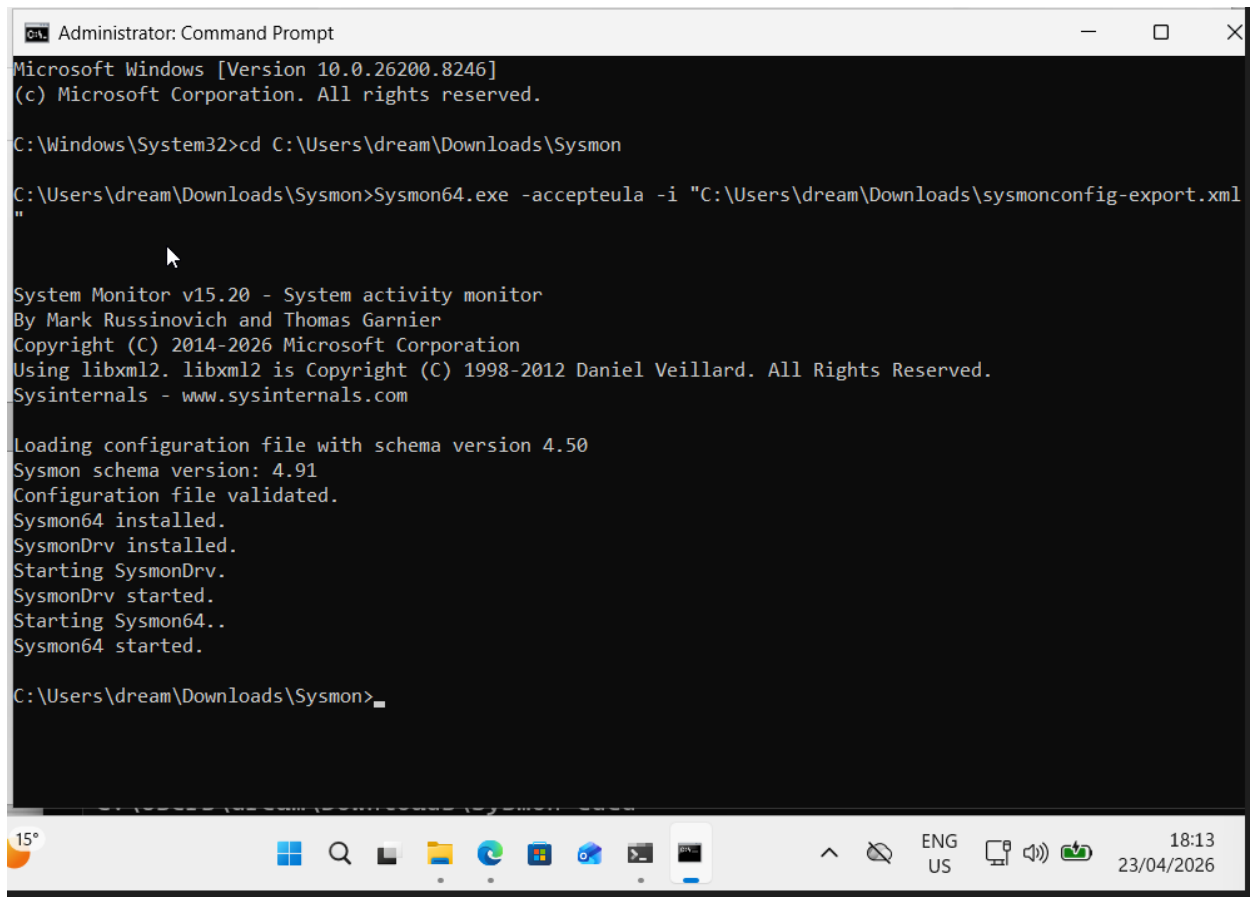
```
  <log_format>eventchannel</log_format>
```

```
</localfile>
```

4. Verification: Confirmed that Sysmons events are appeared in the

Wazuh "Security Events" dashboard.

Sysmon Config-Lab Architecture



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\dream\Downloads\Sysmon

C:\Users\dream\Downloads\Sysmon>Sysmon64.exe -accepteula -i "C:\Users\dream\Downloads\sysmonconfig-export.xml"

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.

C:\Users\dream\Downloads\Sysmon>
```

4. Attack Simulations & Analysis

Technique 1: Pass-the-Hash (T1550.002)

- Description: Using a captured NTLM hash to authenticate as a user without knowing their cleartext password.

- Simulation: Used Atomic Red Team for path

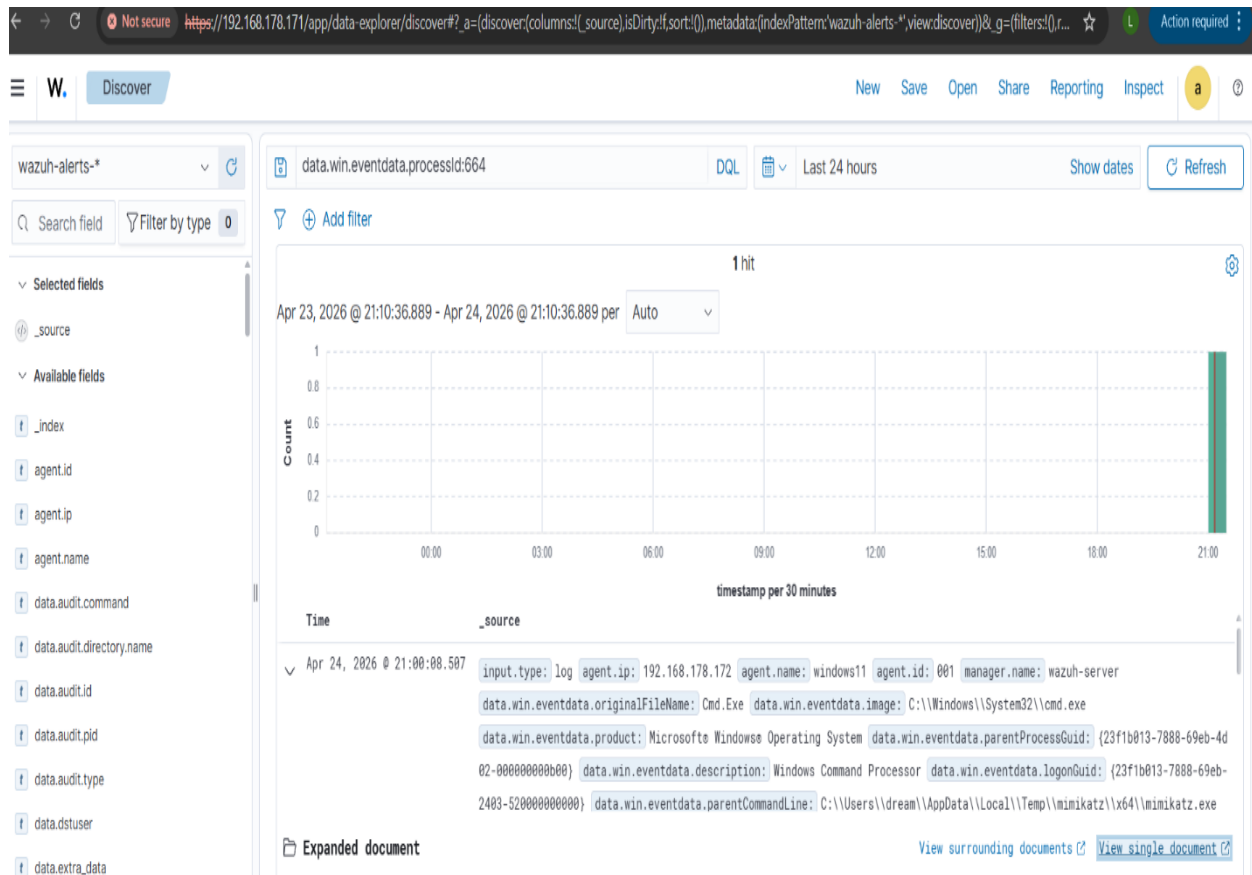
Telemetry Observed:

data.com.eventdata.process ID:664

winlog.event_id:4624 OR data.win.system.eventID:4624

winlog.event_id:4625 OR data.win.system.eventID:4625

winlog.event_id:4672 OR data.win.system.eventID:4672







Technique 2: Kerberoasting (T1558.003)

- Description: Requesting service tickets (TGS) for service accounts with SPNs to crack them offline.
- Simulation: Executed Invoke-Kerberoast from the PowerSploit toolkit.

Telemetry Observed:

`Data.win.system.eventID:4769 AND setspn.exe`

✓ Selected fields

» _source

✓ Available fields

🔍 _index

🔍 agent.id

🔍 agent.ip

🔍 agent.name

🔍 data.extra_data

🔍 data.sca.check.command

🔍 data.sca.check.compliance.cis

🔍 data.sca.check.compliance.cis_csc_v7

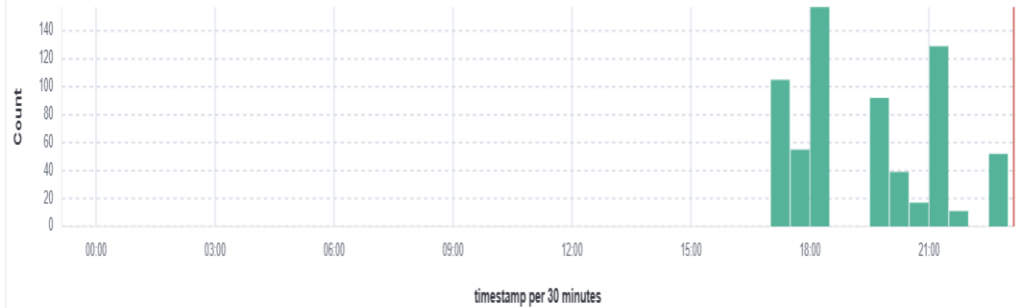
🔍 data.sca.check.compliance.cis_csc_v8

🔍 data.sca.check.compliance.cmmc_v2.0

657 hits

Apr 23, 2026 @ 23:08:18.440 - Apr 24, 2026 @ 23:08:18.440 per

Auto



Time

_source

✓ Apr 24, 2026 @ 22:59:30.609

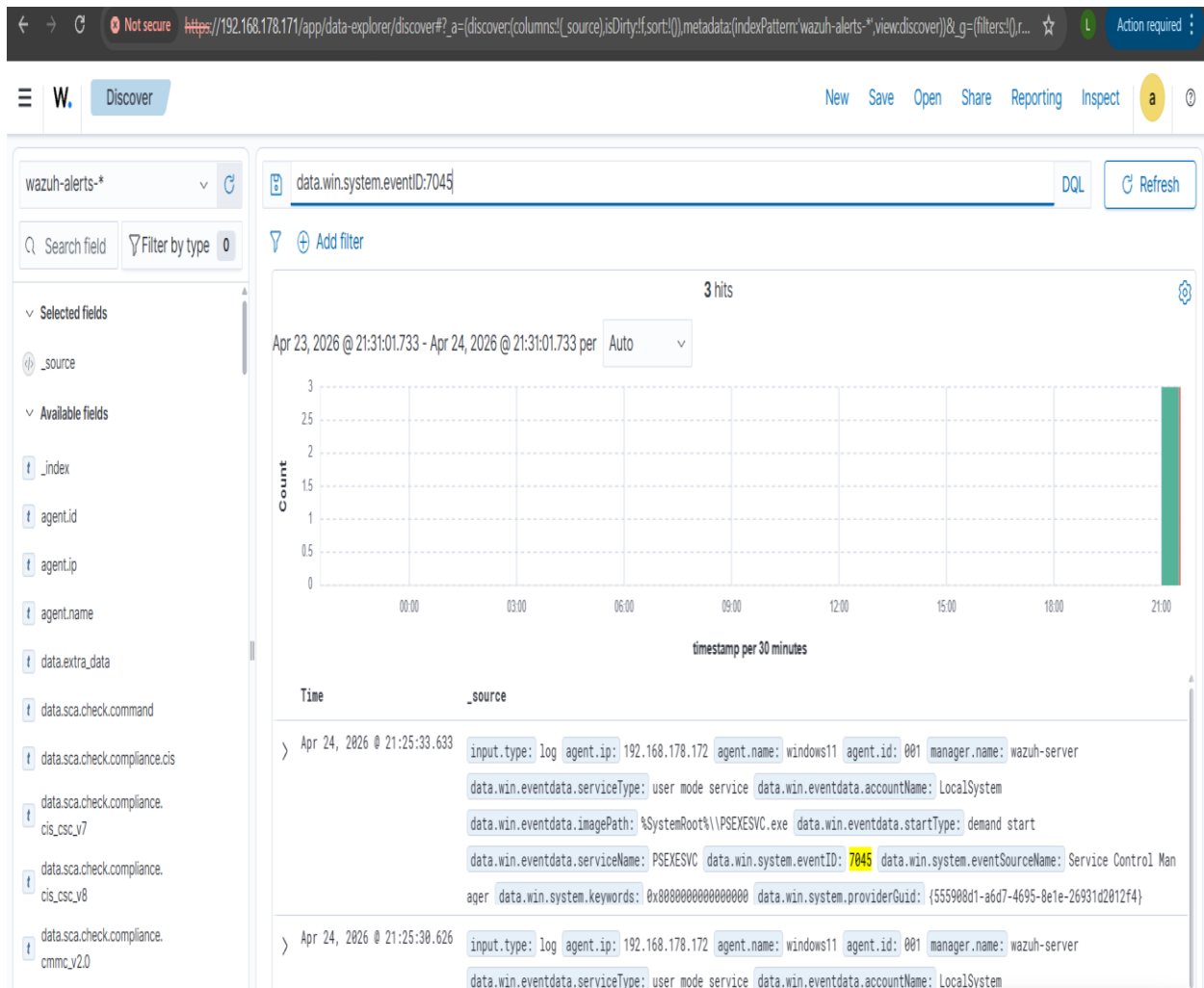
input.type: log agent.ip: 192.168.178.172 agent.name: windows11 agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {23f1b013-8c81-69eb-d203-000000000000} data.win.eventdata.processId: 9156 data.win.eventdata.utcTime: 2026-04-24 15:30:09.572
data.win.eventdata.targetFilename: C:\\Users\\dream\\AppData\\Local\\Temp_PSScriptPolicyTest_bzkw3gaq.t4n.ps1
data.win.eventdata.creationUtcTime: 2026-04-24 15:30:09.572 data.win.eventdata.user: livin\\dream

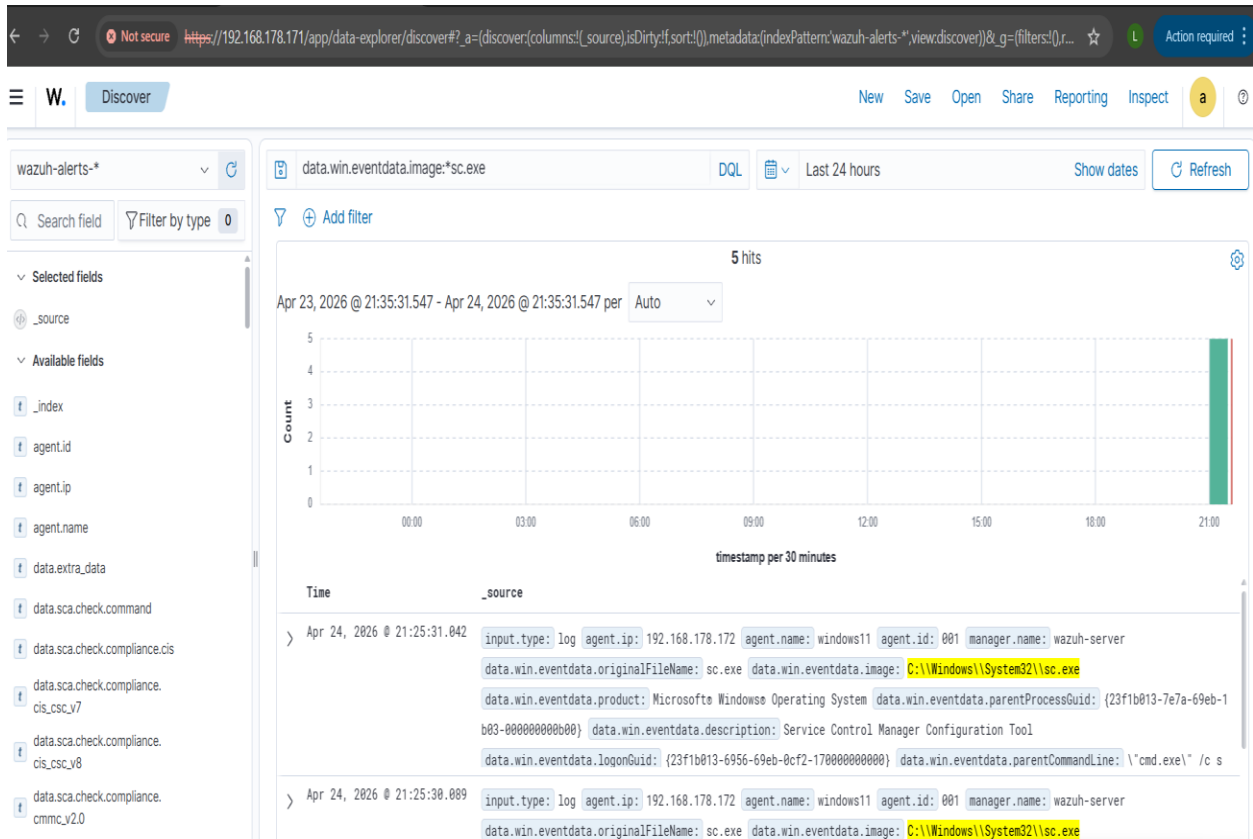
📄 Expanded document

[View surrounding documents](#) [View single document](#)

data.win.eventdata.image:*sc.exe

data.win.system.event.eventID:1





```
Administrator: Windows PowerShell
PS C:\AtomicRedTeam> Invoke-AtomicTest T1569.002 -GetPrereqs
PathToAtomicsFolder = C:\AtomicRedTeam\atomic-red-team\atomics

GetPrereq's for: T1569.002-1 Execute a Command as a Service
No Preqs Defined
GetPrereq's for: T1569.002-2 Use PsExec to execute a command on a remote host
Attempting to satisfy prereq: PsExec tool from Sysinternals must exist in the ExternalPayloads directory
Prereq successfully met: PsExec tool from Sysinternals must exist in the ExternalPayloads directory
GetPrereq's for: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
Attempting to satisfy prereq: PsExec must exist on disk at "C:\AtomicRedTeam\atomic-red-team\atomics\..\ExternalPayloads\PsExec.exe"
Prereq already met: PsExec must exist on disk at "C:\AtomicRedTeam\atomic-red-team\atomics\..\ExternalPayloads\PsExec.exe"
GetPrereq's for: T1569.002-5 Use RemCom to execute a command on a remote host
Attempting to satisfy prereq: RemCom tool must exist on disk in the ExternalPayloads folder
Prereq successfully met: RemCom tool must exist on disk in the ExternalPayloads folder
GetPrereq's for: T1569.002-6 Snake Malware Service Create
No Preqs Defined
GetPrereq's for: T1569.002-7 Modifying ACL of Service Control Manager via SDET
No Preqs Defined
GetPrereq's for: T1569.002-8 Pipe Creation - PsExec Tool Execution From Suspicious Locations
Attempting to satisfy prereq: PsExec tool from Sysinternals must exist in the '\Users\Public\Temp\' directory
Prereq already met: PsExec tool from Sysinternals must exist in the '\Users\Public\Temp\' directory
PS C:\AtomicRedTeam> Invoke-AtomicTest T1569.002 -GetPrereqs
```

```
Administrator: Windows PowerShell
PS C:\AtomicRedTeam> Invoke-AtomicTest T1569.002
PathToAtomicsFolder = C:\AtomicRedTeam\atomic-red-team\atomics

Executing test: T1569.002-1 Execute a Command as a Service
[SC] CreateService SUCCESS
[SC] StartService FAILED 1053:
The service did not respond to the start or control request in a timely fashion.
[SC] DeleteService SUCCESS
Exit code: 0
Done executing test: T1569.002-1 Execute a Command as a Service
Executing test: T1569.002-2 Use PsExec to execute a command on a remote host
The filename, directory name, or volume label syntax is incorrect.
Exit code: 1
Done executing test: T1569.002-2 Use PsExec to execute a command on a remote host
Executing test: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
Remote-to-local symbolic link evaluation is: ENABLED
Remote-to-remote symbolic link evaluation is: ENABLED
The operation completed successfully.
PsExec v2.43 - Execute processes remotely
Copyright (C) 2001-2023 Mark Russinovich
Sysinternals - www.sysinternals.com
'wmic' is not recognized as an internal or external command,
operable program or batch file.
cmd.exe exited with error code 0.
Exit code: 0
Done executing test: T1569.002-4 BlackCat pre-encryption cmds with Lateral Movement
Executing test: T1569.002-5 Use RemCom to execute a command on a remote host
Localhost entered for Target Machine .. Going to RunAs Command
Remote Command Executor
Copyright 2006 The WiseGuyz [ http://talhatariq.wordpress.com ]
Author: Talha Tariq [talha.tariq@gmail.com]
Contributor: Andres Ederra
Local Admin
```

5. Limitations & Future Improvements

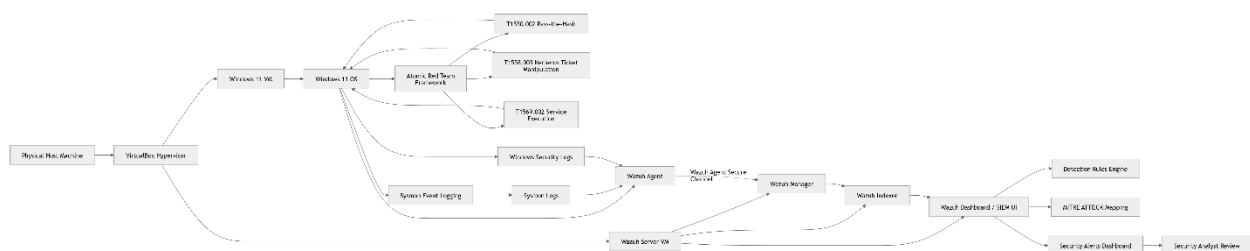
Limitations

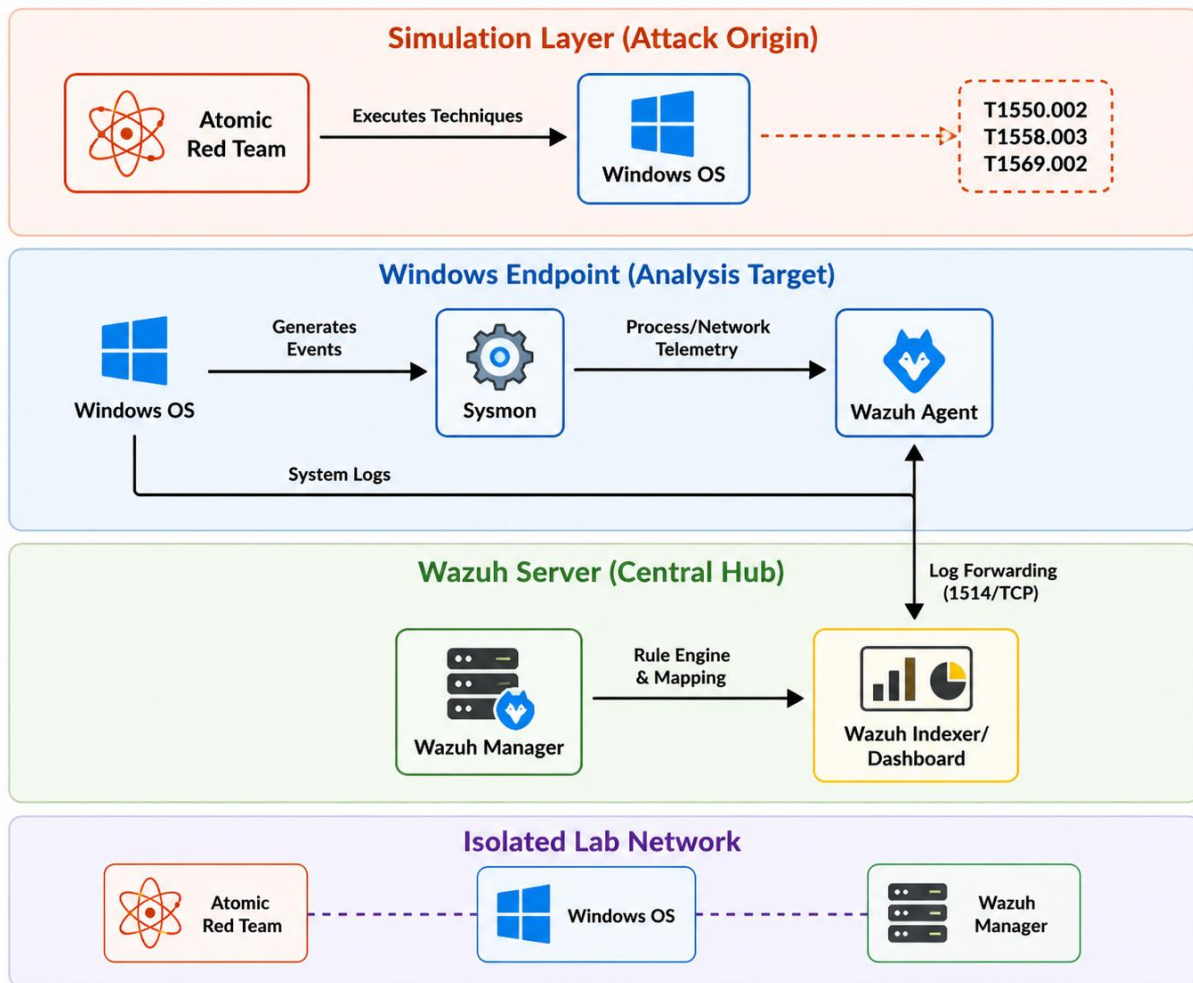
- Configuration Dependency: Detection is only as good as the Sysmon.xml configuration, if a technique doesn't trigger a specific Event ID, Wazuh won't see it.
- Noise: High-volume environment may cause "alert fatigue" without fine-tuned exclusions.

6. Future Improvement

- Automated Response: Configure Wazuh Active Response to isolate the Windows host upon detection of Pass-the-Hash.
- ELK Integration: Enhance visualisation by building custom Kibana dashboards for MITRE ATT&CK mapping.

7. Architectural Diagram





8. Conclusion

The successful implementation of this defensive lab demonstrates the critical synergy between granular endpoint telemetry and centralized security orchestration. By integrating Sysmon with Wazuh, the project achieved high-fidelity visibility into sophisticated

adversary techniques, specifically Pass-the-Hash (T1550.002), Kerberoasting (T1558.003), and Service Execution (T1569.002).

Key takeaways from this challenge include:

- Effective Detection Mapping : The use of Atomic Red Team

allowed for controlled and reproducible simulation, ensuring

The successful implementation of this defensive lab demonstrates the critical synergy between granular endpoint telemetry and centralized security orchestration. By integrating Sysmon with Wazuh, the project achieved high-fidelity visibility into sophisticated adversary techniques, specifically Pass-the-Hash (T1550.002), Kerberoasting (T1558.003), and Service Execution (T1569.002).

Key takeaways from this challenge include:

- Effective Detection Mapping : The use of Atomic Red Team

allowed for controlled and reproducible simulation, ensuring detection logic was accurately mapped to real-world MITRE ATT&CK behaviors.

- **Operational Safety:** All testing was strictly conducted within an isolated lab environment, ensuring no impact on production or shared systems.
- **Technical Analysis:** The project successfully established a pipeline from attack execution to log collection and final analysis within the Wazuh Dashboard.

Ultimately, this project serves as a foundational blueprint for a scalable detection pipelines, meeting the core objectives of clarity, reproducibility, and rigorous analysis.

